



Salimli Ayzek

Data Scientist. 1st-year master's student, SPbSTU, ICCS. Mathematics and Computer Science, Machine Learning and Artificial Intelligence.

- thebloomsbridge - Work
- GitHub
- Hugging Face
- ORCID
- ResearchGate
- eLibrary
- Saint-Petersburg

Версия на русском

Download resume (ENG)

Download this portfolio

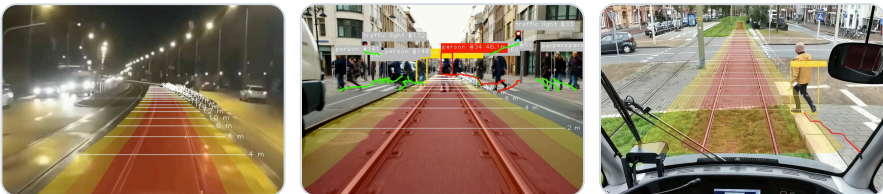
About

- Salimli Ayzek - master's student at SPbSTU (ICCS), majoring in Mathematics and Computer Science: Machine Learning and Artificial Intelligence. I have 1 year and 5 months of experience at The Blooms Bridge: AI systems integration (RAG, computer vision, etc.) and backend development. I primarily position myself as a Data Scientist and work with the ML cycle: data preprocessing and transformation, data analysis, feature selection, model training, hyperparameter tuning, result interpretation, data labeling, LLM training, model building (PyTorch, TensorFlow), deployment (LangChain + LangGraph), and computer vision tasks (YOLO, ResNet, MobileNet, ViT, U-Net, SegFormer, YOLO-seg).
- My professional interests include computer vision, statistical and machine learning algorithms, neural networks, functional programming, optimization methods, statistics, and discrete mathematics.
- Outside work: travel, philosophy (especially how cognitive thinking and foundational ideas connect with computer science), football, and neue deutsche harte.

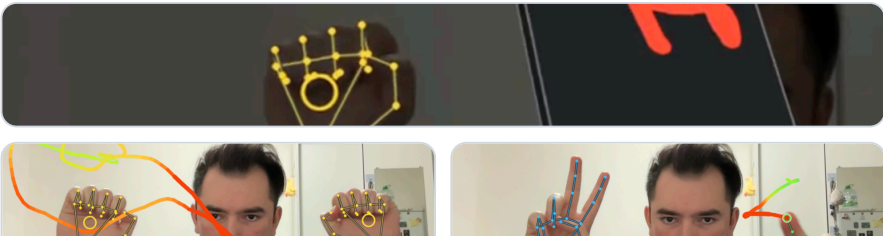
Projects

ML and AI

Object Detection on Dynamic Rol based on Segmentation with Tracking (weights available)

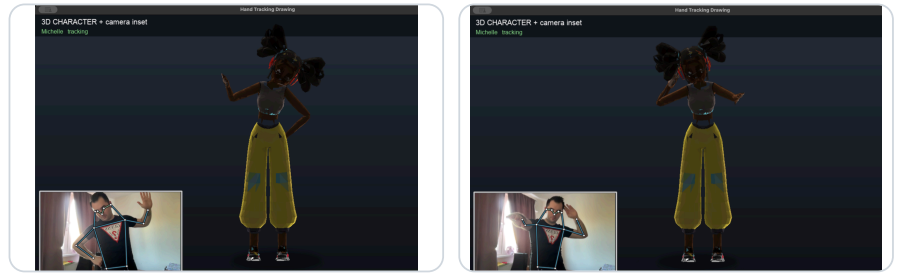


Hand Tracking Drawing (with 3D)

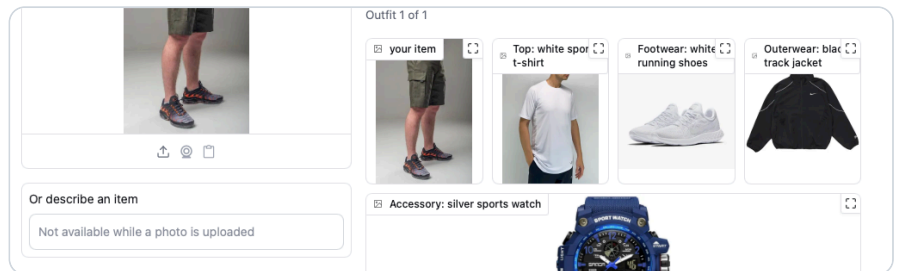




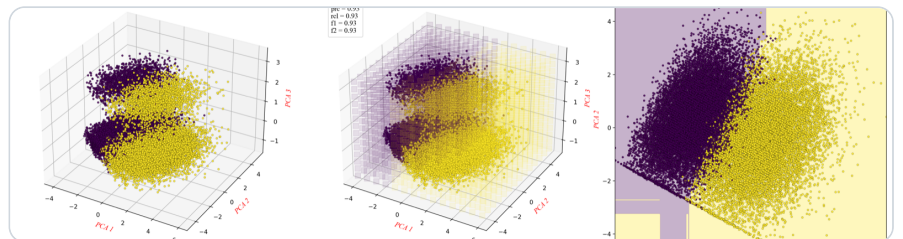
## Pose driven 3D characters



## Vision Fashion Recommender



## Heart Disease Classifier



## Qt MNIST recognizer



## Fashion-MNIST Classification with TinyVGG



## Sarcasmic Text Classification (Transformer NN)

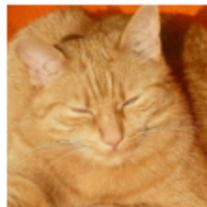
Text: The guy found way to end with Global Hungry, but cant find his keys  
Prediction: Sarcasm  
Confidence: 0.9535

Text: Trump want's to write a law that allows Russians visit USA for free  
Prediction: Not Sarcasm  
Confidence: 0.9781

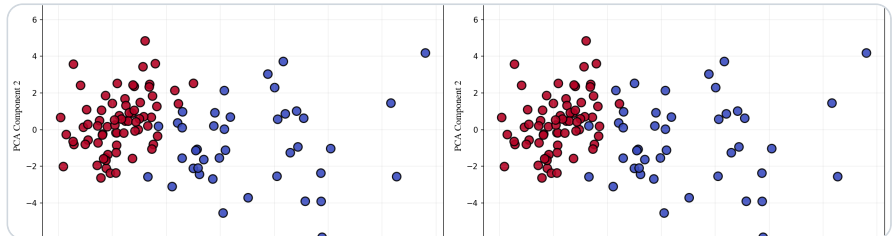
## Leafs Health Classification (GoogLeNet)



## Classification of Cats, Dogs and Wild Animals (CNN)

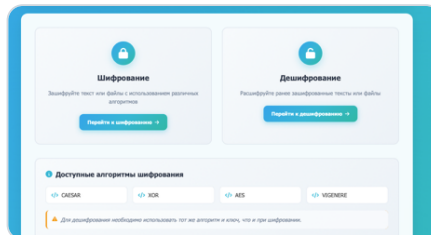


## Breast Cancer Classification



## Web

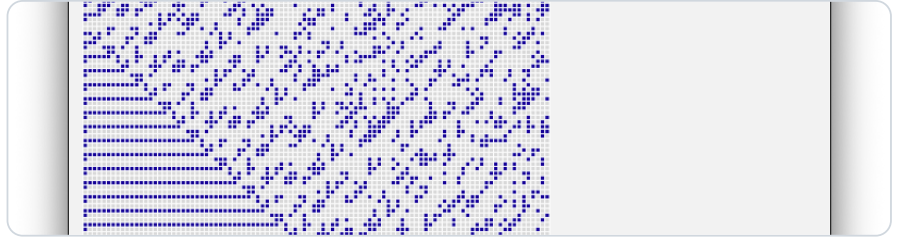
### Data Encryption and Decryption



## Quotes from Movies and Cartoons

## Soft

### Game of Life on CA



[Free MP3 Player iOS: AyzeksSound](#)

[TG-Bot Price Tracker: PriceCheckerBot](#)

[TG-Bot Trip Planner: SpringTripPlannerBot](#)   [HTTP Server on Java](#)

[Card Management Service](#)

## Math

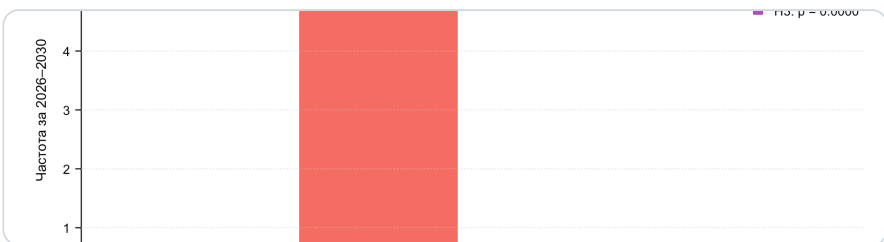
## MathLang Presentations

### Определение

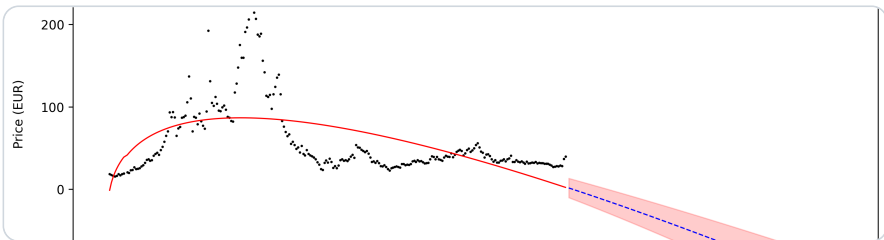
Формально это кортеж состоящий из:

- Множество состояний -  $S$
- Множество входного и выходного алфавита -  $X, Y$
- Начальное состояние -  $s_0$
- Функция переходов -  $\delta: X \times S \rightarrow S$

## Forecasting and Comparison of Demographic Indicators of Russia and Japan



## Regression analysis of the consequences of the reduction in gas export



### Educational Projects

## Education

**School**  
Complete secondary  
Azerbaijan - Baku  
**School №53**  
*Mathematics focus*

**Bachelor**  
Russia - Saint Petersburg  
**Peter the Great St. Petersburg Polytechnic University**  
*Institute of Computer Science and Cybersecurity*  
*Mathematics and Computer Science - AI Systems and Supercomputer Technologies*  
Graduate School of Artificial Intelligence Technologies

**Master's degree**  
1st year  
Russia - Saint Petersburg  
**Peter the Great St. Petersburg Polytechnic University**  
*Institute of Computer Science and Cybersecurity*  
*Mathematics and Computer Science - Machine Learning and Artificial Intelligence*  
Graduate School of Artificial Intelligence Technologies

## Experience

- [The Blooms Bridge](#) - ML Engineer (20.07.2024 - present).

## Articles

- [Construction of the Region of Interest based on segmentation<sup>1</sup>](#)
- [Computer vision for offside detection in football](#)

<sup>1</sup> - Recognized as one of the best articles.

## Diplomas and Certificates



### Skills

| Category                   | Type                        | Technologies / Libraries  |
|----------------------------|-----------------------------|---------------------------|
| Programming Languages      | Multi-paradigm              | C++  Python               |
|                            | Object-Oriented (primarily) | Java                      |
|                            | Functional                  | Haskell                   |
| Data Science & ML (Python) | Data Manipulation           | PANDAS                    |
|                            | Data Visualization          | MATPLOTLIB  SEABORN       |
|                            | Deep Learning (Frameworks)  | PYTORCH  ULTRALYTICS      |
|                            | Machine Learning            | SCIKIT-LEARN              |
|                            | Scientific Computing        | NUMPY  SciPy  statsmodels |
| Quantum                    | -                           | CIRQ                      |

|                            |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computing                  |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Frameworks                 | -                       |  FastAPI  Spring (Core, Boot, Web, Data JPA, Security)                                                                                                                                                                                                                                                                      |
| Databases (DBMS)           | NoSQL                   |  MongoDB                                                                                                                                                                                                                                                                                                                                                                                                       |
|                            | SQL                     |  MySQL  PostgreSQL                                                                                                                                                                                                                                                                                                          |
| Mathematics                | -                       | <ul style="list-style-type: none"> <li>• Discrete Math</li> <li>• Graph Theory</li> <li>• Linear Algebra</li> <li>• Calculus</li> <li>• Mathematical Statistics &amp; Probability Theory</li> <li>• Statistical Learning</li> <li>• Optimization Methods</li> <li>• Quantum Computing</li> </ul>                                                                                                                                                                                                  |
| Message Brokers            | -                       |  RabbitMQ                                                                                                                                                                                                                                                                                                                                                                                                      |
| Containerization & Hosting | -                       |  Docker  Kubernetes  Firebase  Render                                                                                                                 |
| CI/CD & Cloud              | -                       |  Jenkins  Terraform  Ansible<br> HEAT OpenStack Heat  YC |
| Documentation              | Documentation & Reports |  LaTeX  Markdown  Typst                                                                                                                                                                                                            |
|                            | Presentations           |  LaTeX (Beamer)                                                                                                                                                                                                                                                                                                                                                                                              |

## Learning Hub (on progress)

- [Computer Science](#)
- [Mathematics](#)

## Contacts

- [Telegram](#)
- [GitHub profile](#)
- [Gitea profile](#)